Morphohedra (NR-PHMS) — Theory v1: Energy-stable explicit simulation and passive modulation of a gyroscopic topological lattice

Project Morphohedra, Phase 1 working document

30 October 2026

Abstract

We state and prove the four results the Phase-1 implementation relies on: (1) an exact discrete power balance for an explicit scalar-auxiliary-variable leapfrog scheme (ESAV-LF) applied to a two-degree-of-freedom gyroscopic honeycomb lattice with nonlinear on-site potential, Rayleigh-type nonlinear damping and multiple input/output ports; (2) non-negativity of the discrete energy and discrete passivity under the linear CFL condition only; (3) an energy-non-increasing parameter update (ENPU) for time-varying mass, stiffness nonlinearity, gyroscopy and damping; (4) a bound on the energy of energy-gated self-oscillation. Two propositions cover multi-port (MPE) operation and an "energy firewall" coupling to an arbitrary control layer (here a Barkley excitable medium). Every identity was checked symbolically (sympy) and in 50-digit arithmetic (mpmath); the numerical evidence is summarised in the Phase-1 report.

1 Model

Definition 1 (Lattice PHS)**.** *Let N sites carry planar displacements $x_i \in \mathbb{R}^2$, $q = (x_1, \dots, x_N) \in \mathbb{R}^n$, $n = 2N$. Let $M = \mathrm{diag}(m_i I_2)$, $m_i > 0$; $K = k_0 I + \sum_{(i,j)} k_c (e_i - e_j)(e_i - e_j)^\top \otimes n_{ij} n_{ij}^\top \succeq 0$; $U(q) = \sum_{k=1}^n U_k(q_k)$ with $U_k(s) = \frac{\alpha}{4} s^4 + \beta \log\cosh s$, $\alpha, \beta \geq 0$ (so U_k is convex, $U_k \geq 0$, $U_k(0) = 0$); $G = \mathrm{blockdiag}(g_i J)$, $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$; $D(v) = \mathrm{blockdiag}\big((-\mu_i + \eta_i |v_i|^2) I_2\big)$, $\eta_i \geq 0$; $B \in \mathbb{R}^{n \times P}$ (one column per port). The continuous system is $M\ddot{q} = -Kq - \nabla U(q) + G\dot{q} - D(\dot{q})\dot{q} + Bu$, $y = B^\top \dot{q}$, with $\dot{H} = y^\top u - \dot{q}^\top D \dot{q}$ for $H = \frac{1}{2}\dot{q}^\top M \dot{q} + \frac{1}{2} q^\top K q + U(q)$.*

Definition 2 (ESAV-LF)**.** *With $h = 1/f_s$, $c_0 > 0$, $w := q^{n+1} - q^{n-1}$, $\bar{v} := w/(2h)$, $v^{n-1/2} := (q^n - q^{n-1})/h$,*

$$M \frac{q^{n+1} - 2q^n + q^{n-1}}{h^2} = -Kq^n - g^n \odot \frac{\psi^{n+1} + \psi^{n-1}}{2} + (G - D^n)\bar{v} + Bu^n, \tag{1}$$

$$\psi^{n+1} = \psi^{n-1} + g^n \odot w, \qquad g_k^n = \frac{U_k'(q_k^n)}{\sqrt{2U_k(q_k^n) + c_0}}, \qquad D^n := D(v^{n-1/2}).$$

Eliminating ψ^{n+1}, the unknown w solves a block-diagonal system with 2×2 blocks $\left(\begin{smallmatrix} a_x & -c \\ c & a_y \end{smallmatrix}\right)$, $a_\bullet = m_i/h^2 + \frac{1}{2}(g_\bullet^n)^2 + d_i/(2h)$, $c = g_i/(2h)$, $d_i = -\mu_i + \eta_i |v_i^{n-1/2}|^2$, whose determinant $a_x a_y + c^2$ is positive whenever $\mu_i < 2m_i/h$. The scheme is therefore explicit (one 2×2 solve per site, no iteration).

Definition 3 (Discrete energy)**.** $E^{n-1/2} = \frac{1}{2}\|v^{n-1/2}\|_M^2 + \frac{1}{2}(q^n)^\top K q^{n-1} + \frac{1}{4}\sum_k \big((\psi_k^n)^2 + (\psi_k^{n-1})^2\big) - \frac{nc_0}{2}$, *and $\tilde{E} := E + nc_0/2$. At consistency $\psi_k = \sqrt{2U_k + c_0}$, $\frac{1}{4}(\psi^2 + \psi^2) - c_0/2 \approx U_k$.*

2 Results

Theorem 1 (Exact discrete power balance)**.** *For all parameters, states and inputs,* $E^{n+1/2} - E^{n-1/2} = h\,(u^n)^\top y^n - h\,\bar{v}^\top D^n \bar{v}$*, where* $y^n := B^\top \bar{v}$*.*

Proof. Take the inner product of (1) with $w = 2h\bar{v} = h(v^{n+1/2} + v^{n-1/2})$. Since $q^{n+1} - 2q^n + q^{n-1} = h(v^{n+1/2} - v^{n-1/2})$, the left side equals $\|v^{n+1/2}\|_M^2 - \|v^{n-1/2}\|_M^2$. By symmetry of K, $-\langle w, Kq^n\rangle = -\big((q^{n+1})^\top K q^n - (q^n)^\top K q^{n-1}\big)$. From the ψ update, $g^n \odot w = \psi^{n+1} - \psi^{n-1}$, hence $-\frac{1}{2}\langle w, g^n \odot (\psi^{n+1} + \psi^{n-1})\rangle = -\frac{1}{2}\big(\|\psi^{n+1}\|^2 - \|\psi^{n-1}\|^2\big)$. Skew-symmetry gives $\langle w, G\bar{v}\rangle = 2h\,\bar{v}^\top G\bar{v} = 0$. The remaining terms are $-2h\bar{v}^\top D^n \bar{v} + 2h\bar{v}^\top B u^n$. Dividing by 2 and adding and subtracting $\frac{1}{4}\|\psi^n\|^2$ yields the claim. □

Lemma 1. *With* $\bar{q} = (q^n + q^{n-1})/2$ *and* $v = v^{n-1/2}$*:* $(q^n)^\top K q^{n-1} = \bar{q}^\top K \bar{q} - \frac{h^2}{4} v^\top K v$*.*

Proof. $q^n = \bar{q} + \frac{h}{2}v$, $q^{n-1} = \bar{q} - \frac{h}{2}v$; expand. □

Theorem 2 (Non-negativity and passivity under the linear CFL condition)**.** *Let* $\gamma := \frac{h}{2}\omega_{\max} < 1$*,* $\omega_{\max}^2 = \lambda_{\max}(M^{-1}K)$*. Then* $\tilde{E}^{n-1/2} \geq \frac{1}{2}(1-\gamma^2)\|v^{n-1/2}\|_M^2 + \frac{1}{2}\bar{q}^\top K \bar{q} \geq 0$*. If moreover* $\mu_i \leq 0$ *for all* i*, then* $\tilde{E}^{n+1/2} - \tilde{E}^{n-1/2} \leq h\,(u^n)^\top y^n$*; for* $u \equiv 0$ *the energy is non-increasing and, if* $k_0 > 0$*,* $\|q^n\|$ *is bounded by a constant depending only on* $\tilde{E}^{1/2}$*,* k_0 *and* γ*—independently of* α, β, g, η *and of the amplitude.*

Proof. By Lemma 1, $\frac{1}{2}\|v\|_M^2 + \frac{1}{2}(q^n)^\top K q^{n-1} = \frac{1}{2}v^\top(M - \frac{h^2}{4}K)v + \frac{1}{2}\bar{q}^\top K\bar{q}$ and $M - \frac{h^2}{4}K \succeq (1-\gamma^2)M$. The ψ terms are non-negative. If $\mu_i \leq 0$ then $D^n \succeq 0$ and Theorem 1 gives the inequality. Finally $k_0\|\bar{q}\|^2 \leq \bar{q}^\top K\bar{q} \leq 2\tilde{E}$ and $\|v\|_M^2 \leq 2\tilde{E}/(1-\gamma^2)$ bound $q^n = \bar{q} + \frac{h}{2}v$. □

Remark 1. "Unconditional" therefore means: with respect to the nonlinearity, gyroscopy, nonlinear damping and (Theorem 4) gated gain. Only the *linear* CFL condition remains; at the audio parameters used ($f_{\text{unit}} = 1500$ Hz, $f_s = 48$ kHz) $\gamma \approx 0.20$–0.29.

Theorem 3 (ENPU: energy-non-increasing parameter update)**.** *Let* $P \to P'$ *be a parameter change with* P' *admissible* ($\alpha', \beta' \geq 0$*, CFL for* M'*;* K *unchanged). Define* $v_i' = \sqrt{m_i/m_i'}\,v_i$*, the candidate state* $Z(s) = \big(s(\bar{q} + \frac{h}{2}v'),\ s(\bar{q} - \frac{h}{2}v')\big)$ *with* ψ *re-projected* ($\psi_k = \sqrt{2U_k'(\cdot) + c_0}$)*, and* $F(s) := E'(Z(s))$*. Then* $F(s) \leq sF(1)$ *for* $s \in [0,1]$*. Choosing* $s = 1$ *if* $F(1) \leq \max(E_{\text{old}}, 0)$ *and* $s = \max(E_{\text{old}}, 0)/F(1)$ *otherwise gives* $E_{\text{new}} \leq \max(E_{\text{old}}, 0)$*.*

Proof. With re-projection, $\frac{1}{4}(\psi^2 + \psi^2) - c_0/2 = \frac{1}{2}(U_k'(a_k) + U_k'(b_k))$ for the two time levels a, b. By Lemma 1, $F(s) = s^2 A + \frac{1}{2}\sum_k \big(U_k'(sa_k) + U_k'(sb_k)\big)$ with $A = \frac{1}{2}v'^\top(M' - \frac{h^2}{4}K)v' + \frac{1}{2}\bar{q}^\top K\bar{q} \geq 0$. Convexity and $U_k'(0) = 0$ give $U_k'(sa) \leq sU_k'(a)$, so $F(s) \leq s^2A + sB \leq s(A+B) = sF(1)$ for $s \in [0,1]$. If $F(1) > \max(E_{\text{old}}, 0) \geq 0$ the chosen $s \in [0,1)$ gives $F(s) \leq \max(E_{\text{old}}, 0)$. □

Remark 2. (i) Re-projection makes $E \geq 0$ after the update; drift of ψ away from consistency can make $E_{\text{old}} < 0$, in which case $s = 0$ (state reset). This corner case was not hit in the stress tests but must be handled musically (Phase 2: a "keep-ψ" variant that preserves E exactly, valid because K is not modulated). (ii) Any convex U with $U(0) = 0$ is admissible; the theorem does not depend on the specific quartic/log-cosh form.

Theorem 4 (Energy-gated self-oscillation is bounded)**.** *Let* $\mu_i \leq \mu_{\max}$ *with* $\mu_{\max} > 0$*,* $\eta_i \geq 0$*, and apply* $\mu_{\text{eff}} = \mu$ *if* $E^{n-1/2} < E_{\text{cap}}$*, otherwise* $\min(\mu, 0)$*. Let* $\delta = h\mu_{\max}/(m_{\min}(1-\gamma^2)) < 1$*,* $\rho = (1+\delta)/(1-\delta)$*. Then for* $u \equiv 0$ *and with ENPU updates at arbitrary times,* $\tilde{E}^{n+1/2} \leq \max(\tilde{E}^{1/2}, \rho\tilde{E}_{\text{cap}})$*,* $\tilde{E}_{\text{cap}} = E_{\text{cap}} + nc_0/2$*.*

Proof. If the gate is open, $-\bar{v}^\top D^n \bar{v} \le \mu_{\max}\|\bar{v}\|^2 \le \frac{\mu_{\max}}{m_{\min}} \cdot \frac{1}{2}\big(\|v^{n+1/2}\|_M^2 + \|v^{n-1/2}\|_M^2\big) \le \frac{\delta}{h}(\tilde{E}^{n+1/2} + \tilde{E}^{n-1/2})$ by Theorem 2. Theorem 1 gives $(1-\delta)\tilde{E}^{n+1/2} \le (1+\delta)\tilde{E}^{n-1/2} < (1+\delta)\tilde{E}_{\text{cap}}$. If the gate is closed, $\tilde{E}^{n+1/2} \le \tilde{E}^{n-1/2}$. ENPU does not increase E (Theorem 3). Induction completes the proof. □

Proposition 5 (Multi-port / MPE)**.** *With P ports, $h(u^n)^\top y^n = h\sum_k u_k^n y_k^n$: the energy delivered by each voice is metered separately. Terminating any port in a passive load ($u_k = -r_k y_k$, $r_k \ge 0$) or interconnecting with any passive external system through a power-preserving interconnection keeps the composite passive.*

Proof. Immediate from Theorems 1–2; a passive load contributes $-hr_k y_k^2 \le 0$. □

Proposition 6 (Energy firewall)**.** *Let a control layer with arbitrary state z and arbitrary dynamics (possibly driven by audio features and by the lattice output) produce, at control ticks, parameters $P_j = \Phi(z_j)$ clamped to the admissible set $\mathcal{A} = \{m \in [m_{lo}, m_{hi}],\ \alpha, \beta \ge 0,\ \eta \ge 0,\ \mu \le \mu_{\max}\}$, applied via ENPU. Then for $\mu_{\max} \le 0$: $\tilde{E}^n \le \tilde{E}^0 + \sum_{m<n} h(u^m)^\top y^m$; for $\mu_{\max} > 0$ with gating and $u \equiv 0$: $\tilde{E}^n \le \max(\tilde{E}^0, \rho\tilde{E}_{\text{cap}})$.*

Proof. Between ticks Theorems 1, 2 and 4 apply with fixed parameters; at ticks Theorem 3 applies. The only energy channels into the lattice are $u^\top y$ and gated gain; Φ only selects parameters. □

3 Control layer: Barkley medium

On the honeycomb graph Laplacian L: $\dot{u} = \varepsilon^{-1}u(1-u)\big(u - (v+b)/a\big) + D\,Lu$, $\dot{v} = u - v$ (D. Barkley, Physica D 49 (1991) 61–70, doi:10.1016/0167-2789(91)90194-E), explicit Euler with $a = 0.75$, $b = 0.03$, $\varepsilon = 0.02$, $D = 0.5$, $\Delta t = 0.016$, one model time unit $= 1/12\,\text{s}$ (adjustable by sub-stepping). The map Φ: $g_i = g_0(1 - 2\kappa_g u_i)$, $\mu_i = \min(\mu_0 + \kappa_\mu u_i, \mu_{\max})$, $\alpha_i = \alpha_0(1 + \kappa_\alpha u_i)$, all in $\mathcal{A}$, so Proposition 6 applies.

4 Machine verification (summary)

Theorem 1 was verified symbolically (sympy; 1 and 2 sites) and in 50-digit arithmetic (max relative residual $\le 1.0 \times 10^{-50}$ on 1-site, 2×2 and 3×3 flakes). Theorems 2–4 were stress-tested on 8×8 lattices with 4×10^4 randomised trials (see Phase-1 report, gate table).